

Kazi Hassan Shakib

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SUMMARY

PhD Researcher, specializing in quantum cryptography (QC) and post-quantum cryptography (PQC) in securing communications for smart grids and vehicular networks, backed by strong publications and open-source projects, and experienced in both academic research and professional software development.

EDUCATION

Kansas State University <i>Doctorate of Philosophy (All but Dissertation) in Computer Science (GPA: 4.0/4.0)</i>	<i>Manhattan, KS</i> Aug 2027
The University of Alabama <i>Master of Science in Civil Engineering (GPA: 4.0/4.0)</i>	<i>Tuscaloosa, AL</i> Dec 2023
Chittagong University of Engineering and Technology (CUET) <i>Bachelor of Science in Computer Science & Engineering (GPA: 3.33/4.0)</i>	<i>Bangladesh</i> Dec 2018

PROFESSIONAL EXPERIENCE

ISCAAS Lab, Kansas State University <i>Graduate Research Assistant</i>	<i>Manhattan, KS</i> Aug 2024 – Present
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- Led the development of advanced authentication and symmetric key establishment techniques for smart grids leveraging PKC and PQC on resource-constrained devices with C, embedded C, and verification via ProVerif/OCaml.
- Analyzed and mitigated security vulnerabilities in industry standards (ANSI C12.22, IEEE C37.118.2 and IEC 61850-90-5) deployed on Raspberry Pi 500, ESP-8266, STM-32 with C, embedded C.
- Developed QUBO based optimization on Quantum Key Distribution (QKD) and Worked on authentication scheme via Wegman-Carter (WC) in QKD.

CAM Lab, The University of Alabama <i>Graduate Research Assistant</i>	<i>Tuscaloosa, AL</i> Jul 2022 – Jul 2024
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- Evaluated secure VANET communication protocols such as C-V2X, DSRC for improving data integrity and confidentiality with simulation tools like OMNET++, Sumo in C++.
- Identified critical blockchain vulnerabilities employing Quantum Shor’s algorithm, simulated through Qiskit and OMNET++.
- Designed and validated GPS/GNSS spoofing attack detection frameworks built with python and LSTM, contributing to an NSF-funded transportation resiliency project.

IICT, Chittagong University of Engineering and Technology (CUET) <i>Assistant Programmer</i>	<i>Bangladesh</i> Oct 2019 – Jun 2022
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- Automated key university systems and maintained security for applications and servers supporting over 8,000 users, developed with Laravel, PHP.
- Developed and managed web applications for course management, admissions, and 2 international conferences (ICACE 2020, ICMERE 2019) in PHP.
- Conducted research in industrial IoT, machine learning, and data encryption to improve system capabilities applying XGBoost, Random Forest.

City University <i>Full-time Lecturer</i>	<i>Dhaka, Bangladesh</i> Apr 2019 – Nov 2019
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- Taught and mentored over 150 students per semester in undergraduate courses including System Analysis and Design (CSE-325) and Digital System Design (CSE-411).

PROFESSIONAL SKILLS

- Quantum Computing and Post Quantum Cryptography: Qiskit on IBM Quantum Computers, Quantum Error Correction, ML-KEM, Dilithium, Racoon, Gandalf, BB84, E91, CV-QKD etc.
- Machine & Representation Learning: RNN, CNN, QCNN, XGBoost, Random Forest.
- Programming & Tooling: C, C++, PHP, Python, Ocaml, OMNET++, Linux, Git, CUDA (basics), SUMO, Proverif, Laravel.

SELECTED PUBLICATIONS

- Kiesewetter, L., **Shakib, K. H.**, et al. (2023). A holistic review of the current state of research on aircraft design concepts” *Progress in Aerospace Sciences*. **(Cited by 104)**
- **Shakib, K. H.**, Rahman, M., et al. (2025). Impersonation Attack Using Quantum Shor’s Algorithm Against Blockchain-Based Vehicular Ad-Hoc Network. *IEEE Transactions on ITS*. **(Cited by 36)**

AWARDS AND INVOLVEMENT

• President , CS Graduate Student Association (CS-GSA), K-State	2025 – 2026
• Second Place , Poster Presentation, TraCR Annual Conference	May 2024
• Merit Scholarship , Chittagong University of Engineering and Technology (CUET)	2014 – 2018